Input Modeling

Systems Modeling and Simulation
4DV650

Mauro Caporuscio
Department of Computer Science and Media Technology
Mauro.caporuscio@lnu.se

Linnæus University

Why Modeling the input

- Input models provide the driving force for a simulation model
  - E.g., Queueing Systems (Interarrival and Service time), Inventory Systems (Demand, Lead time)

- The quality of the output is no better than the quality of inputs

- Input model development consists of 4 steps:
  1. **Collect data** from the real system
  2. Identify a **probability distribution** to represent the input process
  3. Choose **parameters** for the distribution
  4. **Evaluate** the chosen distribution and parameters for goodness of fit

Data Collection

- One of the biggest tasks in solving a real problem

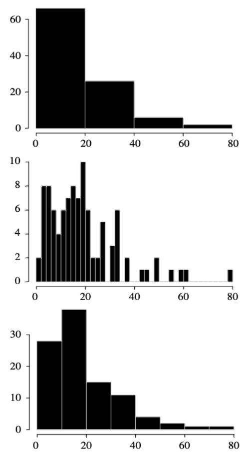
- Suggestions that may enhance and facilitate data collection:
  - **Plan ahead**: begin by a practice or pre-observing session, watch for unusual circumstances
  - **Analyze the data as it is being collected**: check adequacy
  - **Combine homogeneous data sets**, e.g. successive time periods, during the same time period on successive days
  - Be aware of **data censoring**: the quantity is not observed in its entirety, danger of leaving out long process times
  - Check for **relationship between variables**, e.g. build correlation chart
  - Check for **autocorrelation**
  - Collect **input data**, not performance data

1. Histograms
2. Selecting families of distribution
3. Parameter estimation

Identifying the Distribution

Histograms

- A histogram is useful in determining the **shape of a distribution**
- Procedure:
  1. Divide the range of the data into **intervals** (usually of equal width)
  2. Label the horizontal axis to conform to the intervals selected
  3. Find the **frequency of occurrences** within each interval
  4. Plot the frequencies on the vertical axis
- The #intervals depends on the #observations and on the amount of scatter in the data
  - If the intervals are too wide, the histogram will be **coarse**
  - If the intervals are too narrow, the histogram will be **ragged**
  - ➢ #intervals $\approx \sqrt{\#observations}$



Selecting the Family of Distributions

- A family of distributions is selected based on:
  - The context of the input variable
  - Shape of the histogram
- Frequently encountered distributions:
  - **Easier** to analyze: exponential, normal and Poisson
  - **Harder** to analyze: beta, gamma and Weibull (produce wide array of shapes)
- Use the physical basis of the distribution as a guide, for example:
  - Number of *successes* in *n* trials → **Binomial**
  - Number of *independent events* that occur in a fixed amount of time/space → **Poisson**
  - A process that is the *sum of several component processes* → **Normal**
  - *Time* between independent events, or a process time that is memoryless → **Exponential**
  - Processes that are the *sum of several exponentially distributed processes* → **Erlang**
  - *Time to failure* for components → **Weibull**

Quantile-Quantile Plots

- *Q-Q* plot is a useful tool for evaluating distribution fit
- If X is a random variable with cdf F, then the q-quantile of X is the γ such that

$$F(\gamma) = P(X \le \gamma) = q, \qquad \text{for } 0 < q < 1$$

  - When F has an inverse, $\gamma = F^{-1}(q)$

- Algorithm:
  1. Let $\{x_i, i = 1, \ldots, n\}$ be a **sample of data** from X
  2. Order the observations from the **smallest to the largest** and denote them with $\{y_j, j = 1, \ldots, n\}$
     - $y_1 \le y_j \le \cdots \le y_n$, where j denotes the *ranking*
  3. Plot the points $(y_j, F^{-1}((j-0.5)/n))$ *for* $j = 1,\ldots, n$
  4. Check
     - If (plot is "*approximately a straight line*") then F is from an appropriate family of distribution
     - If (the line has slope 1) then F is from an appropriate family of distributions and has appropriate parameter values
     - if (points deviate from a straight line) then the assumed distribution is inappropriate

Parameter Estimation

- If observations in a sample of size *n* are $X_1, X_2, \ldots, X_n$ (discrete or continuous), the **sample mean** and **variance** are:

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \qquad S^2 = \frac{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2}{n-1}$$

- If the data are discrete and have been **grouped** in a frequency distribution:

$$\bar{X} = \frac{\sum_{j=1}^{k} f_j X_j}{n} \qquad S^2 = \frac{\sum_{j=1}^{k} f_j X_j^2 - n\bar{X}^2}{n-1}$$

Where k is the number of distinct values of X and f_j is the observed frequency of the value X_j of X

- When raw data are unavailable (data are grouped into class intervals), the approximate sample mean and variance are:

$$\bar{X} \doteq \frac{\sum_{j=1}^{c} f_j m_j}{n} \qquad S^2 \doteq \frac{\sum_{j=1}^{c} f_j m_j^2 - n\bar{X}^2}{n-1}$$

where f_j is the frequency of X in the interval j, m_j is the midpoint of the interval j, and c is the number of intervals

Suggested Estimators for Distributions Often Used in Simulation

Poisson	α	$\widehat{\alpha} = \bar{X}$
Exponential	λ	$\widehat{\lambda} = \dfrac{1}{\bar{X}}$
Normal	μ, σ^2	$\widehat{\mu} = \bar{X}$ $\widehat{\sigma}^2 = S^2$

Selecting Input models without Data

- Some possible sources to obtain information about the process are:
  - **Engineering data**: often product or process has performance ratings provided by the manufacturer or company rules specify time or production standards
    - e.g., a laser printer can produce 8 pages/minute
  - **Expert option**: people who are experienced with the process (or similar processes), can provide useful insights
    - e.g., optimistic, pessimistic and most-likely times
  - **Physical or conventional limitations**: physical limits on performance, limits or bounds that narrow the range of the input process
    - e.g. computer data entry cannot be faster than a person can type
  - **The nature of the process**
    - For modeling a Number of *successes* in *n* trials use the Binomial Distribution (see slide 6)
- The uniform, **triangular** and beta distributions are often used as input models.

Materials (MyMoodle)

- Reading
  - 04 [Lecture] Input Modeling - Reading.pdf

- Exercises (with solutions):
  - 04 [Lecture] Input Modeling - Exercises.pdf
  - 04 [Lecture] Input Modeling - Exercises.pdf